

C Language

Curriculum

UNIT 1: Getting Started with C

Lessons:

- History and Features of C
- Installing Compiler (GCC)
- Writing First Program (Hello World)
- Structure of C Program
- Compilation and Execution Process
- Comments and Basic Syntax

Module Assessment:

- Basic MCQ Test
- Write and execute simple C program

UNIT 2: Variables and Data Types

Lessons:

- Data Types (int, float, double, char)
- Type Modifiers (short, long, signed, unsigned)
- Variables and Naming Rules
- Constants (#define, const)
- Type Casting
- Scope of Variables

Module Assessment:

- Output prediction
- Simple arithmetic programs

UNIT 3: Operators

Lessons:

- Arithmetic Operators
- Assignment Operators
- Increment and Decrement Operators
- Relational Operators
- Logical Operators
- Bitwise Operators
- Operator Precedence and Associativity

Module Assessment:

- Expression evaluation
- Simple calculator program

UNIT 4: Strings and Text Processing (Basics)

Lessons:

- Character Arrays
- String Declaration and Initialization
- String Input and Output
- String Functions (strlen, strcpy, strcmp, strcat)
- Basic String Manipulation

Module Assessment:

- String handling programs

UNIT 5: User Input Handling

Lessons:

- scanf usage
- Format Specifiers

- Multiple Inputs
- Input Buffer Issues
- Safe Input Techniques

Module Assessment:

- User-based calculation programs

UNIT 6: Control Flow

Lessons:

- if Statement
- if-else and Nested if
- switch-case
- while Loop
- do-while Loop
- for Loop
- break and continue
- goto statement

Module Assessment:

- Pattern printing
- Number-based logic programs

UNIT 7: Advanced Strings and Character Arrays

Lessons:

- String Traversal using Loops
- Reverse String
- Palindrome Check
- Count Characters and Words
- Custom String Functions

Module Assessment:

- Text analyzer program

UNIT 8: Code Debugging and Error Handling

Lessons:

- Syntax Errors
- Runtime Errors
- Logical Errors
- Debugging Techniques
- Dry Run and Tracing
- Common Programming Mistakes

Module Assessment:

- Debug and correct given programs

UNIT 9: Functions

Lessons:

- Function Declaration and Definition
- Function Parameters
- Call by Value
- Return Types
- Recursion
- Storage Classes (auto, static, extern, register)

Module Assessment:

- Recursive programs
- Function-based modular programs

UNIT 10: Arrays

Lessons:

- One Dimensional Arrays
- Array Input and Output
- Searching Techniques
- Sorting (Bubble Sort)
- Two Dimensional Arrays (Matrix)
- Matrix Operations

Module Assessment:

- Matrix addition
- Max/Min element finder

UNIT 11: Pointers

Lessons:

- Introduction to Pointers
- Address and Dereference Operators
- Pointer Arithmetic
- Pointers with Arrays
- Pointers with Functions
- Dynamic Memory Allocation (malloc, calloc, free)

Module Assessment:

- Swap using pointers
- Dynamic array creation

UNIT 12: Structures and Unions

Lessons:

- Defining Structure
- Accessing Structure Members

- Array of Structures
- Nested Structures
- Passing Structure to Functions
- typedef
- Union Basics

Module Assessment:

- Student record management program

UNIT 13: File I/O Basics

Lessons:

- File Opening Modes
- fopen and fclose
- fprintf and fscanf
- fread and fwrite
- File Error Handling
- CSV File Processing

Module Assessment:

- File-based data storage program

UNIT 14: Beginner Level Projects

Projects:

- Todo List Application
- Quiz Game
- Contact Manager
- Word Counter

Module Assessment:

- Mini project submission

- Code explanation

UNIT 15: Intermediate Level Projects

Projects:

- Inventory Management System
- Student Database Manager
- File Encryption Tool
- Banking System (File Based)

Module Assessment:

- Complete working project
- Documentation and viva